

sample construction by varying minimum firm size thresholds. We consider alternative task-level weights to construct our exposure measures. We estimate weighted regressions that give greater weight to larger firms and occupations to ensure our results reflect typical worker experiences. Finally, we relax the assumption of linear treatment effects by categorizing firms and occupations into exposure quartiles. These analyses consistently support our main findings.

These robustness checks notwithstanding, we acknowledge the possibility that other factors differentially affected exposed and less exposed firms or occupations around the time of ChatGPT’s introduction. For instance, if firms anticipating AI disruption had already begun adjusting their workforce composition before November 2022, or if other technological or economic shocks coincidentally affected high-exposure entities differently, our estimates could be biased. Additionally, our exposure measures capture potential susceptibility to LLM disruption but may not perfectly predict actual adoption patterns, which likely vary with firm-specific factors such as technological capabilities, management quality, and financial resources.

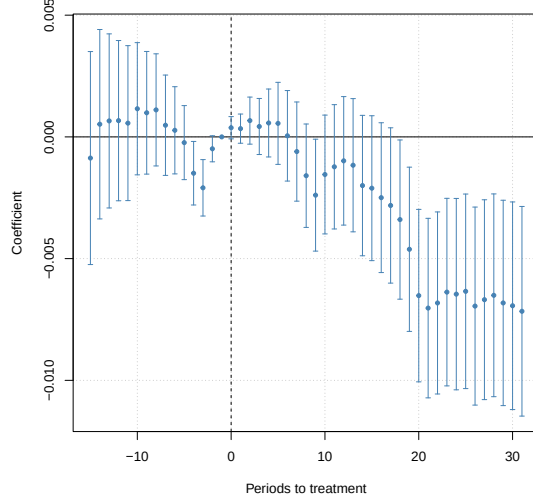
## 4 Results

This section presents our main empirical findings on the effects of LLM exposure on UK labor market outcomes. We organize the analysis in three parts. First, we examine how employment stocks, workforce composition, and compensation evolve within firms with different levels of LLM exposure. Second, we investigate occupation-level hiring patterns through job posting data. Third, we explore within-firm recruitment strategies by combining firm-level exposure measures with their occupation-specific hiring patterns. Throughout, we present both event study specifications and static difference-in-differences estimates.

### 4.1 Firm-Level Employment Effects

Figure 1 presents the event study plots for the effect of LLM exposure on the total number of employees in UK firms. We find that highly exposed firms start experiencing a statistically significant decline in total employment beginning approximately 18 months after the introduction of ChatGPT, suggesting that exposed firms reduce headcount in response to this new technology, but that it takes time to

Figure 1: Effect of LLM exposure on firm-level jobs



*Notes:* This figure plots event study estimates of the effect of LLM exposure on the total number of employees in UK firms (log). Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. Vertical line indicates November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

show up in aggregate employment statistics. High and low LLM exposure exhibit similar employment trajectories prior to November 2022, which provides support for the validity of the parallel trends assumption. When averaged over the entire post-treatment period (Table 3), a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment, although the aggregate effect is only marginally significant due to the delayed onset of impacts.

We next examine how these aggregate effects vary across different types of workers. Figure 2 decomposes the effects between high- and low-seniority employees. Panel A shows junior positions decline immediately and statistically significantly, while Panel B demonstrates that senior positions initially remain stable before declining gradually, with the decrease only becoming significant after 20 months. Panel C shows that the timing difference in the emergence of effects leads to a temporary shift toward senior employees, although the compositional change attenuates over time, with the seniority distribution gradually reverting toward pre-treatment levels by the end of our observation period. All three panels show parallel pre-trends.

Table 3: Effect of LLM exposure on firm-level job outcomes

|                       | Employees          | Low seniority       | High seniority    | Frac. low seniority  | Av. comp.            |
|-----------------------|--------------------|---------------------|-------------------|----------------------|----------------------|
| LLM exposure x Post   | -0.003*<br>(0.002) | -0.004**<br>(0.002) | -0.002<br>(0.002) | -0.001**<br>(0.0003) | 106.313*<br>(59.029) |
| Firm fixed effects    | Yes                | Yes                 | Yes               | Yes                  | Yes                  |
| Industry-by-period FE | Yes                | Yes                 | Yes               | Yes                  | Yes                  |
| Observations          | 1,268,013          | 1,267,955           | 1,266,331         | 1,267,955            | 1,268,013            |
| Adj. R-squared        | 0.983              | 0.983               | 0.981             | 0.979                | 0.978                |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, averaged over the post-treatment period. *Employees* is the total number of jobs in the firm. *Low seniority* and *High seniority* denote the number of positions in seniority levels 1–3 and 4–7, respectively. *Low seniority (%)* is the share of jobs that are low-seniority. *Av. comp.* is the mean annual salary plus bonuses. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level and reported in parentheses. Significance is denoted by \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

The regression results in Table 3 show, averaged over the entire post-treatment period, a significant 0.4 percent reduction in junior employees per standard deviation increase in exposure, and a statistically insignificant reduction in senior employees. The fraction of junior employees goes down by 0.1 percentage points per standard deviation of LLM exposure ( $p = 0.028$ ).

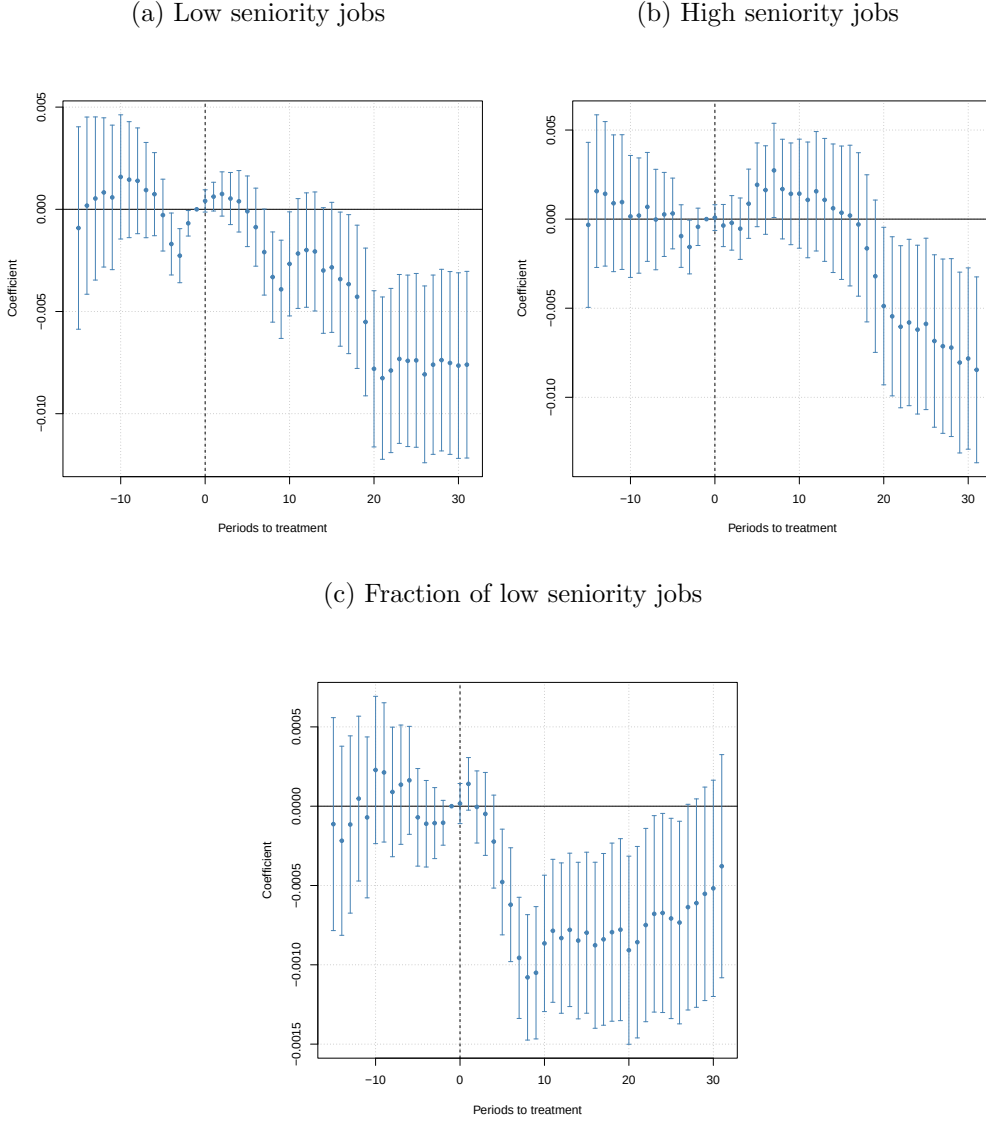
The observed compositional shift towards more senior roles has direct implications for compensation patterns. Figure 3 shows that average compensation rises modestly in highly exposed firms following the introduction of ChatGPT, with a one standard deviation increase in exposure associated with a £106 increase in average compensation, although the overall estimate is only marginally significant.

## 4.2 Job Posting Dynamics

### 4.2.1 Firm-Level Postings

While employment stocks adjust slowly, job postings provide a real-time window into firms’ evolving labor demand. We therefore examine how firms’ hiring intentions respond to LLM exposure by analyzing whether exposed firms post fewer vacancies overall and whether they adjust the composition of roles they seek to fill. This posting data captures the extensive margin of recruitment activity and reveals forward-looking decisions that precede the employment stock changes documented above. By tracking the same firms across both datasets, we can directly observe

Figure 2: Effect of LLM exposure on firm-level seniority of jobs



*Notes:* This figure plots event study estimates of the effect of LLM exposure on the seniority composition of jobs within UK firms. Panel A shows effects on the total number of low-seniority positions (seniority levels 1–3), panel B on the total number of high-seniority positions (levels 4–7), and panel C on the fraction of jobs that are low-seniority. Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

how hiring intentions translate into gradual changes in employment.